## HORIZON EUROPE — MARIE SKŁODOWSKA-CURIE ACTIONS

### Postdoctoral Fellowships 2026 — European Fellowship

Call: HORIZON-MSCA-2026-PF-01-01 | Type of Action: HORIZON-TMA-MSCA-PF-EF

# DairyTwin

## A Minimum-Viable, Explainable Digital Twin for Dairy Cows on Commercially Available Farm Data

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|---|---|
| **Proposal acronym** | DairyTwin |
| **Full title** | A Minimum-Viable, Explainable Digital Twin for Dairy Cows on Commercially Available Farm Data |
| **Topic / Type of Action** | HORIZON-MSCA-2026-PF / HORIZON-TMA-MSCA-PF-EF (European Postdoctoral Fellowship) |
| **Researcher (applicant)** | Ing. Jan Saro, Ph.D., MBA — Czech Republic |
| **Sending institution** | Czech University of Life Sciences Prague & Institute of Information Theory and Automation (ÚTIA), Czech Academy of Sciences |
| **Beneficiary (host)** | Technische Universität Dresden — ScaDS.AI Dresden/Leipzig (Center for Scalable Data Analytics and AI), Germany |
| **Supervisor (host)** | Prof. Dr.-Ing. Wolfgang Lehner — Chair of Database Systems, Faculty of Computer Science, TU Dresden; co-founder of ScaDS.AI Dresden/Leipzig |
| **Duration** | 12 months |
| **Mobility direction** | Czech Republic → EU host |
| **Keywords** | digital twin; precision dairy; decision support; explainable AI; Markov decision processes; open science; animal welfare |

A one-page online project summary, with downloadable supporting materials, is available at jenasin.github.io/msca-dairytwin.

### List of abbreviations

| | |
|---|---|
| AMS | Automatic Milking System |
| DIM | Days In Milk |
| ETL | Extract-Transform-Load (data pipeline) |
| GLM | Generalized Linear Model |
| HMM | Hidden Markov Model |
| MDP | Markov Decision Process |
| MSCA-PF | Marie Skłodowska-Curie Actions — Postdoctoral Fellowships |
| RL | Reinforcement Learning |
| SHAP | SHapley Additive exPlanations |
| WP | Work Package |
| XAI | Explainable Artificial Intelligence |

## Excellence

### Quality and pertinence of the project's R&I objectives and the extent to which they are ambitious and go beyond the state of the art

**What a digital twin is, and how this project translates the idea to a cow.** The concept of a *digital twin* was introduced by Michael Grieves in 2002–2003 in a Product Lifecycle Management course at the University of Michigan and formalised in his 2014 white paper (Grieves 2014; Grieves & Vickers 2017) as a tightly coupled triple: a *physical entity*, a *virtual representation* of that entity, and a *bidirectional data link* between the two such that the virtual side stays in step with the physical side over its entire lifecycle. The construct was first deployed at scale by NASA for aerospace and spacecraft monitoring (Glaessgen & Stargel 2012) and matured through manufacturing and Industry 4.0 (Tao et al. 2019). The transfer to *biological* and *agricultural* subjects is more recent and incomplete: Neethirajan & Kemp (2021) introduced the term to livestock farming as a vision paper, Pylianidis et al. (2021) systematised the agricultural case in *Computers and Electronics in Agriculture*, and Defraeye et al. (2021) demonstrated digital twins for perishable supply chains. None of these works delivers an engineering-grade twin of an individual dairy cow that runs on data farms already export and produces a daily, ranked, explainable worklist — the artefact DairyTwin sets out to build.

**Translating the triple to a dairy cow.** In DairyTwin the *physical entity* is the individual cow in the herd; the *virtual representation* is a latent state vector $\mathbf{s}_t = (\mathbf{s}_t^{\text{prod}}, \mathbf{s}_t^{\text{pert}}, \mathbf{s}_t^{\text{fert}}, \mathbf{s}_t^{\text{risk}})$ updated daily; the *data link* is the nightly bundle of parlour and AMS logs, milk-recording records, activity–rumination collar data, weather and management events that the farm already exports. The twin's output is not another raw score on top of the existing dashboards — it is a constrained, ranked, sentence-explained daily worklist that closes the loop between sensing and acting. This is, deliberately, a *minimum-viable* twin: it does not attempt to replace the farmer's judgement, model rumen biochemistry, or run continuously on every signal at sub-second granularity. It runs once per day, on data the farm already has, and outputs only what can be acted on the same morning.

**How the twin will be built (12-month plan).** The work proceeds in five overlapping work packages (Section 3): an ETL pipeline and a formal cow-state schema over open datasets (WP1, M1–3); a hybrid state estimator combining a probabilistic state-space layer (Bayesian filter / HMM / MDP) with an ML residual layer (gradient boosting / penalised GLM) (WP2, M3–6); a constrained herd-prioritisation policy with integer, greedy and contextual-bandit variants under explicit operator-time and treatment-budget constraints (WP3, M5–9); a farmer-facing explainability layer anchored to each cow's own baseline, with reason codes, counterfactual sentences and uncertainty banding (WP4, M7–11); and an MIT-licensed open prototype with a reproducible benchmark over Zenodo, Agroscope, INRAE and Teagasc datasets (WP5, M10–12). Each WP has its own deliverables, validation gate and risk register (Section 3); each is sized so that even partial completion of any single WP leaves the others independently publishable.

**Transferable beyond the dairy cow.** The state-space schema is designed to be *species-portable*, not cow-specific. The four-component decomposition (production / perturbation / fertility / risk) and the data envelope (per-animal yield or output, activity, weather, management events) generalise directly to other production species on a typical mixed European farm: *dairy goats and sheep* (parlour data analogous to cows, smaller herd sizes, similar mastitis epidemiology); *beef cattle* (weight-gain trajectories in place of milk yield, oestrus and fertility tracking near-identical); *fattening pigs* (feeding-station logs in place of milk records, similar perturbation and risk states); and, with a different sensor mix, *laying hens* (egg-production trajectories, environmental sensors). Only the per-species observation model, the lactation/growth/laying curve, and the disease ontology change; the latent-state semantics, the constrained prioritiser, and the cow-own-baseline explanation framework remain unchanged. This portability is built in from the schema

(WP1) and is an explicit acceptance criterion for the open prototype (WP5), so that follow-up groups — in Czech goat herds, in Central European beef and pig operations, in pasture-based New Zealand or Irish flocks — can re-instantiate the twin for their species at low marginal cost.

European dairy is among the most data-rich and most decision-poor sectors of agriculture. A 2026 cow generates thousands of digital events per day, yet many small and medium farmers still decide which cow to look at first using memory and end-of-month spreadsheets. The literature is strong in isolated cow-level monitoring (Rutten et al. 2013; Besler et al. 2024) and moderately strong in cow-level risk models (Hooker et al. 2025; Liu et al. 2025), but weak in the integration of latent-state estimation, herd-level prioritisation under realistic resource constraints, and farmer-facing explanation into a single, reproducible, openly licensed workflow.

**DairyTwin addresses precisely this gap.** The project designs, implements and validates a hybrid digital twin of an individual dairy cow that (i) runs on data EU dairy farms already export every day, (ii) models each cow as an explicit latent state with four components (production, perturbation, fertility, risk trajectory), (iii) prioritises cows under operator-time and treatment-budget constraints, (iv) explains each daily recommendation in a single sentence against the cow's own baseline, and (v) is validated on retrospective and open datasets so the artefact can be reused by other groups.

**Specific R&I objectives.**

- **O1** Define a minimum-viable digital-twin schema for a single cow and a herd, with formal state-space semantics, over commercially available data.
- **O2** Develop a hybrid state estimator (probabilistic state-space layer + machine-learning residual layer) that outperforms one-shot baselines on lead time and precision-at-top-$k$, not on AUROC alone.
- **O3** Build a constrained herd-prioritisation policy (integer / greedy / contextual-bandit variants) under explicit operator constraints.
- **O4** Deliver a farmer-facing explainability layer based on cow-own-baseline reason codes, counterfactual sentences, and uncertainty banding.
- **O5** Release an open, MIT-licensed prototype with a reproducible benchmark over Zenodo, Agroscope, INRAE and Teagasc datasets.
- **O6** Validate the twin retrospectively against alert burden, lead time, calibration, precision-at-top-$k$, and simulated workload — the metrics that match how a vet or farmer actually uses such a system.

**What is new in this approach.** The state of the art divides into three islands — cow-level predictive models, herd-level economic simulators, and farmer-facing explanation work — each largely developed in isolation. DairyTwin is the first system that fuses all three into a single workflow validated end-to-end against *operator-relevant* metrics. Concretely, four design choices distinguish DairyTwin from any published dairy decision-support system to date:

- **(N1) Cow as an explicit latent state, updated daily and carried forward**, rather than as the subject of one-shot daily classifications. The state $(\mathrm{s}^{\text{prod}}, \mathrm{s}^{\text{pert}}, \mathrm{s}^{\text{fert}}, \mathrm{s}^{\text{risk}})$ is propagated through time and feeds the herd-level decision directly.
- **(N2) Herd prioritisation under explicit operator constraints**, treating veterinary time, treatment budget and inspection slots as first-class inputs to a constrained allocation policy — not as post-hoc filters of a flat risk list.
- **(N3) Explanations anchored to the cow's own baseline**, not to the population ("rumination $-18\%$ vs. 7-day mean" instead of "feature importance = 0.27"). This is what a farmer can actually verify on the next visit to the stall.
- **(N4) End-to-end evaluation on operator-relevant metrics** — lead time, precision@top-$k$, alert burden, simulated workload, calibration — rather than on AUROC alone.

**Relevance to the MSCA Work Programme.** The project contributes to several EU priorities concurrently: the Farm-to-Fork strategy (welfare-first decision support, reduced antimicrobial use); the European Data Strategy and the Common European Agricultural Data Space (open prototypes over open and consent-cleared data); the European Green Deal (decision support that improves animal welfare and herd longevity, lowering methane intensity per litre of milk); and Horizon Europe's emphasis on open, reproducible, citizen-trustworthy AI.

## Soundness of the methodology, including underlying concepts, interdisciplinarity, gender dimension and open-science practices

**Methodological backbone.** Each cow is represented as a latent state $\mathbf{s}_t = (\mathbf{s}_t^{\text{prod}}, \mathbf{s}_t^{\text{pert}}, \mathbf{s}_t^{\text{fert}}, \mathbf{s}_t^{\text{risk}})$ updated daily from the multi-source signal $\mathbf{o}_t = (\text{milk}, \text{AMS}, \text{collar}, \text{weather}, \text{events})$. A probabilistic state-space layer (Bayesian filter / HMM / MDP) produces calibrated daily scores and uncertainty bands; an ML residual layer (gradient boosting / penalised GLM) captures non-linear deviations the state-space layer does not. Herd-level prioritisation is posed as a constrained allocation problem solved by integer / greedy / contextual-bandit variants and evaluated against a rule-based baseline.

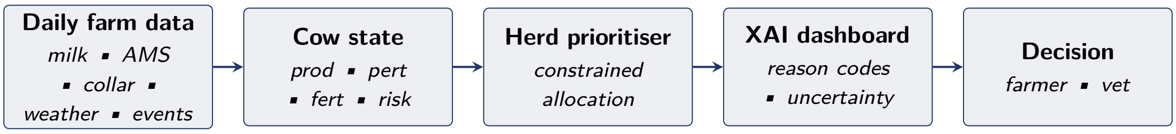



*Figure 1.* DairyTwin architecture: data farms already export every day flow through a cow-level state estimator, a herd-level constrained prioritiser, and a farmer-facing explanation layer, ending in a ranked worklist that the farmer or vet can verify and act on.

**Interdisciplinarity.** DairyTwin sits explicitly at the intersection of three communities: precision dairy and animal welfare (the host-side strength), operations research and decision science (the applicant's Czech base at CULS and ÚTIA), and applied data and machine learning (the host-side methodological depth). The work plan distributes deliverables across all three from M1.

**Gender dimension and other diversity aspects.** Although the primary research subject is dairy cattle, the project incorporates a gender-aware design in three places: (i) the user interface, language and explanation style of the prototype are co-designed with male and female extension agents (WP4); (ii) case studies and interviews used in the co-design panel are balanced by gender, age and farm size; (iii) authorship, visibility and conference presentations created by the project are allocated transparently. Where data on farm workers are touched, they are anonymised and aggregated; no individual operators or workers are identifiable in any released artefact.

**Open science practices.** The project commits to open science from M1. All code is released under an MIT licence on a public Git platform with reproducible notebooks and a CI-tested benchmark. All datasets used are either openly licensed (Zenodo, Agroscope, INRAE, Teagasc, data.gov.ie) or processed under written agreements that allow anonymised derived data to be published. Pre-prints are deposited on arXiv / Zenodo on acceptance. Article-processing charges are budgeted so that publications are gold open-access. A Data Management Plan and an Open Science Plan are maintained throughout the fellowship and reviewed jointly with the supervisor at six-month intervals (D1.2 and D5.3).

**Research integrity, ethics and animal-welfare framing.** Animal welfare is treated as a first objective, not a constraint to be relaxed for yield. Following the 2021 "Farmers Bill of Rights" commentary and the 2025 update on dairy data governance, no commercial farm data are re-exported; any commercial data, where touched, are processed under written agreement and anonymised at source. The project does not require ethics approval for animal experimentation, as the work is retrospective and observational; an ethics self-assessment is included in Part B2.

### Quality of the supervision, training and of the two-way transfer of knowledge

The fellowship is structured as a two-way exchange between the host group and the fellow. The host provides depth in scalable data analytics on heterogeneous time-series, data-stream management, calibration of model outputs against operational queries, and reproducibility infrastructure — the technical core of Prof. Wolfgang Lehner's Database Systems Group at TU Dresden and of ScaDS.AI Dresden/Leipzig. The fellow brings stochastic processes, Markov decision modelling, decision-support engineering for dairy, and ten years of professional software-engineering experience in critical infrastructures.

**Two-way knowledge transfer table.**

| From host → fellow | From fellow → host |
|---|---|
| Host's specialty (precision dairy, scalable data analytics, or livestock welfare); access to commercial farm data; mentoring on EU-grant proposal writing; integration into PhD training programme. | Stochastic processes & Markov modelling; decision-support engineering; deployment-grade software practice; non-specialist communication track record; Czech–Central European agricultural network. |

**Training plan.** Embedded co-supervised work package; research integrity & open-science training; transferable skills (grant writing, scientific communication, supervision of master's students); mobility (one secondment at month 12–13, two major conferences, one training school). Weekly research meetings with the supervisor; quarterly review with an external advisor.

### Quality and appropriateness of the researcher's professional experience, competences and skills

Ing. Saro is an unusually well-positioned candidate for an MSCA-PF in this domain.

**Education and academic record.** Ph.D. in Systems Engineering, Czech University of Life Sciences Prague (defended 2025); MBA, Newton University Prague (2022); M.Sc. in Open Informatics — Computer Engineering, Czech Technical University in Prague (2015); B.Sc. in Open Informatics — Computer Systems, Czech Technical University in Prague (2012).

**Current positions.** Assistant Professor, Department of Systems Engineering, FEM, CULS Prague (2025–); Junior Researcher, Department of Adaptive Systems, ÚTIA, Czech Academy of Sciences (2025–).

**Doctoral publications and awards.** Five peer-reviewed publications on dairy decision support, predictive modelling and livestock emissions (full record on ORCID 0009-0001-4355-5271):

1. Saro, J. *et al.* Decision support systems in dairy cows farming: a 20-year scoping review of characteristics, applications, and future challenges. *Czech Journal of Animal Science* (2026). DOI: 10.17221/40/2026-CJAS.
2. Saro, J. *et al.* Regional patterns and cluster analysis of agricultural methane emissions in the EU-27 countries. *Czech Journal of Animal Science* (2025). DOI: 10.17221/26/2025-CJAS.
3. Saro, J. *et al.* Discrete Homogeneous and Non-Homogeneous Markov Chains Enhance Predictive Modelling for Dairy Cow Diseases. *Animals* **14**, 2542 (2024). DOI: 10.3390/ani14172542. *Dean's First Prize for the best publication by a Ph.D. student at FEM CULS 2024.*
4. Saro, J. *et al.* A decision support system for herd health management for dairy farms. *Czech Journal of Animal Science* (2024). DOI: 10.17221/178/2024-CJAS.
5. Saro, J. *et al.* A decision support system based on disease scoring enables dairy farmers to proactively improve herd health. *Czech Journal of Animal Science* (2024). DOI: 10.17221/53/2024-CJAS.

**Industry and applied experience.** A decade of professional software-engineering experience in critical infrastructures, including the European EESSI social-security exchange (S&T CZ), public-sector citizenfacing systems (ISFG), Avast / Gen Digital, and a long-running freelance practice (50+ delivered web systems). This combination of mathematical rigour and deployment skill is exactly what an MSCA postdoc in an applied AI domain requires.

**Public-engagement track record.** National finalist of the Falling Walls Lab Czech Republic 2024 with the deliberately non-technical talk *Breaking the Wall of the Dairy Farmer AI Assistant*. This is one of the dimensions the MSCA programme is explicitly designed to reward.

## Impact

**EU dairy at a glance — why this matters.** The EU produces approximately **155 million tonnes of cow milk** per year (Eurostat, 2023) from about **20 million dairy cows** on more than **1.4 million dairy farms**. Subclinical mastitis is estimated to cost the EU dairy sector on the order of **€1.5 billion per year** (literature estimates, Hogeveen et al. and successors), with an average per-case cost of **€240–300**. Even a conservative 5 % reduction in unnecessary inspections and antimicrobial treatments achieved through better daily prioritisation would represent an EU-level saving on the order of **€75–100 M per year** and a measurable contribution to the EU's antimicrobial-stewardship and Farm-to-Fork targets. DairyTwin is built precisely to unlock that scale-level efficiency without removing the human from the decision.

### Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to their skills development

DairyTwin is structured so that the fellow leaves the fellowship with three durable assets: (i) a hardened open prototype that can serve as the technological backbone of a follow-up Horizon Europe project, a national grant (Czech GAČR, Czech TAČR), or a deployment in a Czech / EU agritech start-up; (ii) three peer-reviewed publications across dairy science, decision support and AI methodology (target venues: *Journal of Dairy Science*, *Computers and Electronics in Agriculture*, *IEEE Access*); and (iii) a transatlantic and pan-EU scientific network seeded by an open seminar at the host, a return seminar in Prague, and at least two major-conference presentations.

**Career trajectory.** The fellow's trajectory — industry software engineer → Ph.D. in applied AI for dairy → MSCA postdoc with a deployable open prototype — is consistent with one of the outcomes the MSCA programme is explicitly designed to enable: a research-academic profile in an applied domain where deployment skill is in short supply. The fellow plans to use the fellowship as the launchpad for an ERC Starting Grant application within 24 months of the fellowship ending.

**Reintegration.** On return, the fellow will integrate the prototype into the Systems Engineering and Decision Support master's curriculum at CULS (approximately 80 students per year), co-supervise at least one master's thesis and one Ph.D. project that extend the twin to Czech farm data, and seek a follow-up grant.

### Quality of the proposed measures to exploit and disseminate the project results

**Exploitation.** The open prototype is released under an MIT licence and made available to Czech and EU extension services, farmer associations and agritech partners free of charge. A commercially deployable variant of the prototype is positioned as a documented project output that Czech dairy farms, cooperatives or agritech start-ups (in the EU or jointly with U.S. partners) can license or co-develop. The open reference implementation and the commercial-grade deployment are complementary, not competing.

**Dissemination.** Three peer-reviewed papers (target venues above), at least two conference presentations

(EAAP Annual Meeting, ECPLF), a co-organised workshop on explainable decision support in dairy at the host institution, contribution to the relevant EU data-space community group. All publications are preprinted; all code and benchmarks are openly archived (Zenodo DOI).

**IPR and FAIR.** All outputs follow FAIR principles. Where commercial collaboration produces patentable IP, ownership is governed by a written agreement that explicitly preserves the open reference implementation; no commercial agreement will close the open core.

**Stakeholder engagement.** Structured engagement with two extension organisations (one Czech, one in the host country) and two commercial dairies of varying size, used as a co-design panel for the explanation layer.

### Quality of the proposed measures to communicate the project activities to different target audiences

A two-track communication plan runs through the fellowship: (i) scientific communication (papers, talks, code releases, presentations at academic conferences) and (ii) public communication. The latter includes a non-technical web summary and a short video for Czech farmer-facing media; a Czech-language popular-science write-up at the end of the project; participation in the host country's annual researchers' night; and a six-monthly blog log on the project landing page (jenasin.github.io/msca-dairytwin, to be extended for MSCA). The fellow has prior experience in non-specialist communication, recognised at Falling Walls Lab 2024.

**Audiences and channels.**

| **Audience** | **Channel** | **Message** |
|---|---|---|
| Scientific community | Peer-reviewed papers, ECPLF / EAAP talks, preprints | State-of-the-art digital twin for dairy with end-to-end open evaluation. |
| Industry and start-ups | Open-source release, GitHub, dedicated landing page | A deployable prototype that can be productised. |
| Farmers / extension | Czech-language popular-science article, short video, demo at extension fair | A daily worklist tool that explains itself. |
| General public | Researchers' Night, social media short-form | Why an AI assistant for cows is a small instrument of welfare and fairness. |

## Implementation

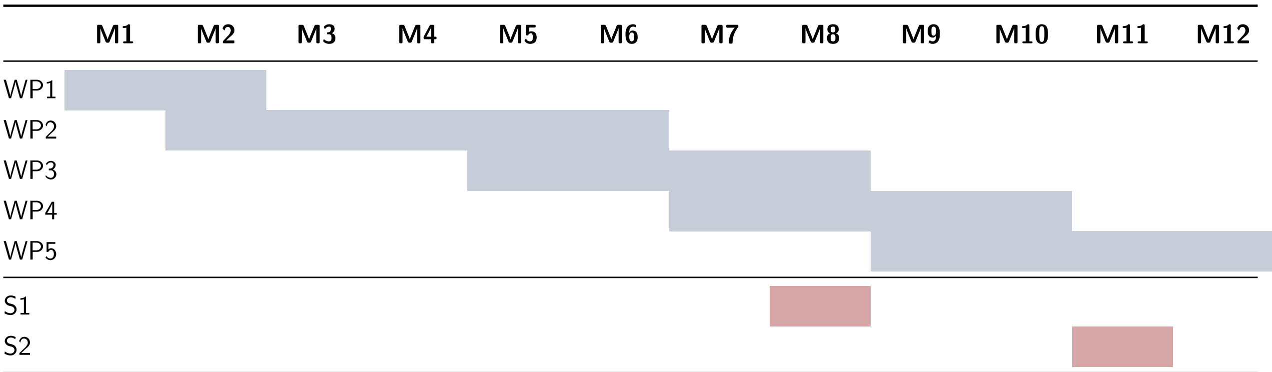
### Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

DairyTwin is organised into five work packages over a focused 12-month plan (within the 12–24 month MSCA-PF range). The 12-month design is deliberately tight: deliverables are scoped to what is realistically achievable, and stretch goals are explicitly tagged in the risk register. One short secondment (**S1**) and one training-school placement (**S2**) are built into the work plan, both shown in the Gantt below.

*Work packages and effort allocation*

| WP | Title | Description | Months | Effort (PM) |
|---|---|---|---|---|
| WP1 | Scope, data inventory, schema | Lock cow-state representation, ETL and quality-control pipeline; secure public-dataset access; data management and ethics self-assessment. | M1–M2 | 2 |
| WP2 | Cow-level state estimator | Probabilistic state-space + ML residual layer; baseline benchmark; leave-one-farm-out validation. | M2–M6 | 4 |
| WP3 | Herd-level prioritisation | Constrained allocation policy (integer / greedy / bandit); what-if simulator. | M5–M8 | 2 |
| WP4 | Explainability & dashboards | Cow-own-baseline reason codes, counterfactual sentences, uncertainty banding, dashboard, co-design. | M7–M10 | 2 |
| WP5 | Validation, dissemination & exploitation | Retrospective evaluation against operator-relevant metrics; papers; open release; communication; return-phase planning. | M9–M12 | 2 |
| | | | **Total person-months** | **12** |

*Gantt overview (12-month plan)*




*Legend:* work-package activity secondment / training-school. **S1**: 1-month secondment to a complementary EU dairy-science partner at the end of WP3 (M8); **S2**: 1-week training school / workshop on responsible AI and open science (M11).

*Deliverables and milestones*

| ID | Title | Description | Due |
|---|---|---|---|
| D1.1 | Data dictionary & twin schema | Cow-state schema, data envelope, quality-control checklist. | M2 |
| D1.2 | Data Management Plan | FAIR plan + Open Science Plan; updated biannually. | M2 |
| D2.1 | State estimator (open) | Probabilistic + ML hybrid; reproducible notebooks; baseline metrics. | M6 |
| D3.1 | Herd prioritiser | Constrained policy + what-if simulator. | M8 |
| D4.1 | Explainability layer & dashboard | SHAP, reason codes, counterfactuals, uncertainty banding. | M10 |
| D5.1 | Public benchmark & design paper | Reproducible benchmark + first peer-reviewed paper submitted. | M11 |
| D5.2 | Final report & open release | Final report; open-source release; Czech popular-science article; return-phase plan. | M12 |
| D5.3 | Updated Open Science Plan | Final version of the Open Science Plan reviewed with supervisor. | M12 |
| S1 | Secondment to a complementary EU dairy-science partner | 1-month placement at the end of WP3 to validate the prioritisation layer in a different farm and data context. | M8 |
| S2 | Training school on responsible AI and open science | 1-week training school / workshop, aligned with MSCA transferable-skills training. | M11 |
| MS1 | Schema and pipeline locked | End of WP1. | M2 |
| MS2 | State estimator beats baselines on precision@top-$k$ and lead time | End of WP2. | M6 |
| MS3 | Closed-loop demonstrator (state $\rightarrow$ priority $\rightarrow$ reason) | End of WP4. | M10 |
| MS4 | Retrospective validation accepted by independent reviewers | End of WP5. | M12 |

*Risk register*

| Risk | Mitigation | Impact × Likelihood |
|---|---|---|
| Commercial farm data not accessible in time | Designed to fall back to open datasets (Zenodo / Agroscope / INRAE / Teagasc / data.gov.ie) without losing deliverables. | Med × Med |
| Vendor-aggregated collar data hides preprocessing | Reason-code layer uses cow-own baselines, robust to vendor smoothing. | Med × High |
| Domain shift between farms | Leave-one-farm-out evaluation built into the pipeline from M3. | Med × Med |
| Alert fatigue in the prototype | "No action" is a first-class output category; uncertainty banding is enforced. | High × Med |
| Scope creep | 12-month plan with tight deliverables; stretch goals explicitly tagged and deferred beyond the fellowship if needed. | Med × Med |
| Personal continuity (illness, family) | Pre-agreed remote-work clauses with host; quarterly reviews; full remote-capable workflow from M1. | Low × Low |

### Quality and capacity of the host institution(s) and participating organisations, including hosting arrangements

The host institution is **Technische Universität Dresden**, specifically the **ScaDS.AI Dresden/Leipzig — Center for Scalable Data Analytics and Artificial Intelligence**. ScaDS.AI is one of six German National Competence Centers for AI, jointly operated by TU Dresden and Leipzig University, with a remit covering scalable methodology, AI for natural sciences, and trustworthy and explainable AI — the three pillars on which DairyTwin is built.

**Why this host.** The Dresden site of ScaDS.AI brings depth in scalable data analytics, ML methodology, and federated and trustworthy AI; this complements the applicant's strengths in stochastic processes, Markov decision modelling, and decision-support engineering, and is precisely the technical complement that turns a cow-level prototype into an explainable, reproducible, deployable system. The host group's role in the German national AI ecosystem also opens immediate channels for dissemination, FAIR archiving, and follow-up funding through Horizon Europe and German national calls.

**Hosting arrangements.** TU Dresden provides: dedicated office space and computing resources (including access to GPU compute through ScaDS.AI infrastructure); integration into ScaDS.AI's group seminar and the TU Dresden PhD training programme; co-supervision and weekly research meetings; a one-month secondment to a complementary EU group at the end of WP3; visa, accommodation and integration support through the TU Dresden Welcome Centre; and full administrative support during the fellowship. A signed Letter of Commitment from TU Dresden accompanies the submission. The capacity of the host is detailed further in Part B2.

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*End of Part B1. Part B2 (CV, Capacity of Participating Organisation, Ethics Self-Assessment, Declarations) is provided in a separate file. Letters of Commitment are uploaded separately. Online project summary at jenasin.github.io/msca-dairytwin.*